



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
13-09-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

PAPER-1

Time : 3 Hours

Maximum Marks : 192

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

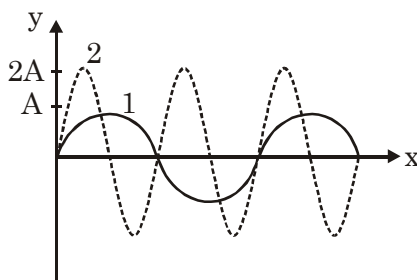
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 40)

- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

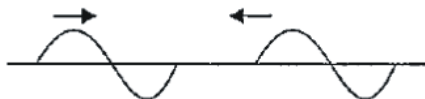
1. The graph shown here depicts displacement versus position graph for two progressive sine waves. The graph shown by dark line represents wave-1 and that shown by dotted line shown wave-2. Both the waves are in the same medium. Choose the **CORRECT** statement(s) :-



- (A) The average power carried by wave-2 is 16 times the average power carried by wave-1.
- (B) The energy contained in one wavelength of 2 is 8 times that of wave-1
- (C) The time period of wave-2 is twice of wave-1
- (D) The maximum velocity of the particles of medium in case of wave-2 is 4 times that of wave-1

Space for Rough Work

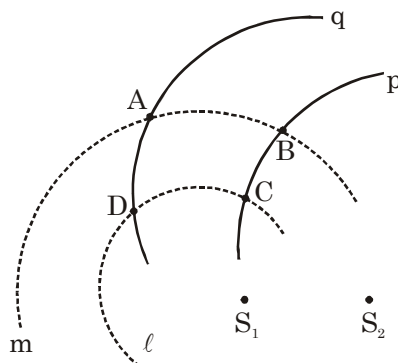
2. A string oscillates in its third harmonic with a frequency of 195 Hz. The tension in the string is 338 N, and its length is 2 m. Which of the following options is **CORRECT**?
- (A) If the string is fixed at one end only, its mass is 2.5 gm.
 (B) If the string is fixed at one end only, its mass is 5 gm.
 (C) If the string is fixed at both the ends, its mass is 10 gm.
 (D) If the string is fixed at both the ends, its mass is 20 gm.
3. Two identical sinusoidal pulses move in opposite directions on a stretched string. Kinetic energy of each pulse is k . Choose the **CORRECT** statement(s)



- (A) The total mechanical energy of each pulse at the moment shown is $2k$.
 (B) At the instant they overlap completely kinetic energy of the resulting pulse is $2k$.
 (C) At the instant they overlap completely the potential energy of resulting pulse is $4k$.
 (D) The mechanical energy at the moment shown is uniformly distributed along the pulse.

Space for Rough Work

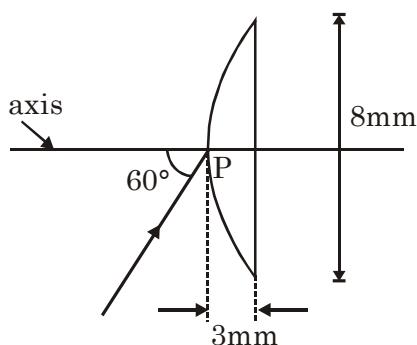
4. In the figure shown. S_1 and S_2 are two identical point sources of sound which are coherent 180° out of phase. Taking S_1 as centre, two circular arcs ℓ and m of radii 1m and 2m are drawn. Taking S_2 as centres, two circular arcs p and q are drawn having radii 2m and 4m respectively. It is given that wavelength of wave produced by each source is 4.0 m . Which of the following statement is/are **CORRECT**.



- (A) A will record maximum intensity.
 (B) C will record maximum intensity
 (C) B will record minimum intensity.
 (D) D will record minimum intensity

Space for Rough Work

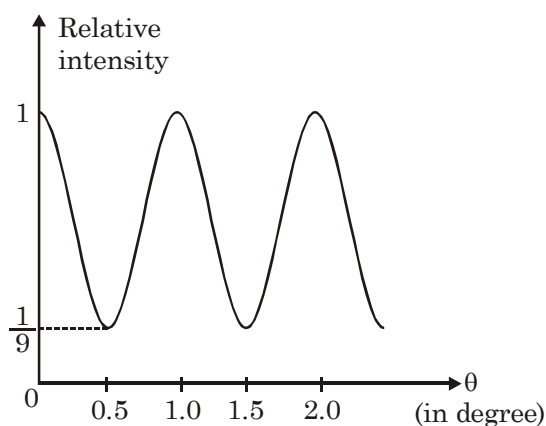
5. A plane convex lens has aperture diameter of 8mm and thickness of the lens at the centre is 3mm. The refractive index of the material of the lens is $\sqrt{3}$. A light ray is incident at mid point P of the curved surface at an angle of incidence of 60° . Which of the following is/are **CORRECT** ? (Take : $2\sqrt{3}(\sqrt{3}-1) = 2.5$)



- (A) Angular deviation suffered by the ray as it passes through the lens is 30° .
- (B) Radius of curvature of the curved surface is $\frac{25}{6}$ mm.
- (C) Radius of curvature of the curved surface is $\frac{25}{3}$ mm.
- (D) Lateral shift in the path of the ray as it passes through the lens is $\sqrt{3}$ mm.

Space for Rough Work

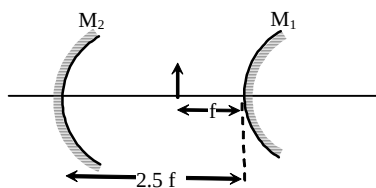
6. In young's double slit experiment relative intensity at point on the screen may be defined as ratio of intensity at that point to the maximum intensity on the screen. Light of wavelength $7500 \text{ \AA}$ passing through a double slit. Produces interference pattern of relative intensity variation as shown in figure θ on horizontal axis represents the angular position of a point on the screen. Which of the following is/are **CORRECT** :-



- (A) Separation 'd' between the slit is 0.02 mm.
 (B) Separation 'd' between the slit is 0.04 mm.
 (C) Ratio of amplitudes of two waves producing interference pattern on the screen is 2 : 1.
 (D) Ratio of amplitudes of two waves producing interference pattern on the screen is 3 : 1.

Space for Rough Work

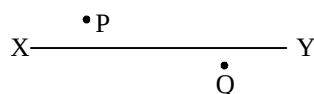
7. A source of sound, producing a sinusoidal wave, is moving uniformly towards an observer at a velocity of 20 m/s. The observer is moving away from the source at a constant velocity of 10 m/s. Frequency of the source is 200 Hz and speed of sound in air is of 340 m/s.
- (A) In an interval of 10 second, the eardrums of the observer will sense maximum change in pressure 2062 number of times.
- (B) Apparent wavelength of sound for the observer is 1.7 m
- (C) Apparent wavelength of sound for observer is 1.6 m.
- (D) In an interval of 10 second, the eardrums of the observer will sense maximum change in pressure 4125 number of times.
8. The two mirrors M_1 & M_2 of focal length f each are placed coaxially as shown. Consider first reflection from M_1 & second from M_2 . Then which of the following are **CORRECT** :



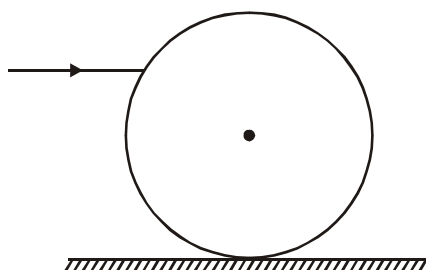
- (A) The image after two reflection coincides with object
- (B) Image after two reflection is inverted and its size is one fourth of object
- (C) Image after two reflection is real.
- (D) image after two reflection is virtual

Space for Rough Work

9. Two points P and Q lie on either side of an axis XY as shown in the figure. It is desired to produce an image of luminous point object kept at P at Q using a spherical mirror, with XY as the optic axis. The mirror must be -



- (A) converging (B) diverging
(C) positioned to the left of P (D) positioned to the right of Q
10. A cylindrical glass rod of radius 0.1 m and refractive index $\sqrt{3}$ lies on a horizontal plane mirror. A horizontal ray of light moving perpendicular to the axis of the rod is incident on it at height H from the mirror so that it leaves the rod at a height of 0.1 m above the plane mirror. At centre to centre distance of d, a second similar rod, parallel to the first, is placed on the mirror, such that the emergent ray from the second rod is in line with the incident ray on the first rod, then



- (A) value of H can be 0.186 m (B) value of H can be 0.086 m
(C) value of d can be 0.32 m (D) value of d can be 0.12 m

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

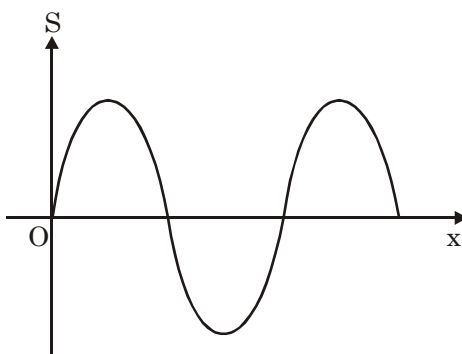
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	●	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A long string of mass per unit length 0.2 Kg/m is stretched to a tension 500N. A transverse wave of amplitude $A = 10$ mm and $\lambda = 0.5$ m is traveling on this string. This string is joined to another string of same cross section and mass per unit length 0.8 Kg/m. The fraction of the average power carried by the wave in the first string that is transmitted to the second string is K. Find K.
2. A longitudinal harmonic wave is travelling along positive x direction. The amplitude, wavelength and frequency of the wave are 8.0×10^{-3} m, 12 cm and 6800 Hz respectively. The displacement(s) versus position graph for particles on the x-axis at an instant of time has been shown in figure. Find the separation (in cm) at the instant shown, between the particles which were originally at $x_1 = 1$ cm and $x_2 = 3$ cm.



Space for Rough Work

3. Standing waves are produced by the superposition of two waves of the form

$$y_1 = 15 \sin (3\pi t - 5\pi x) \text{ cm};$$

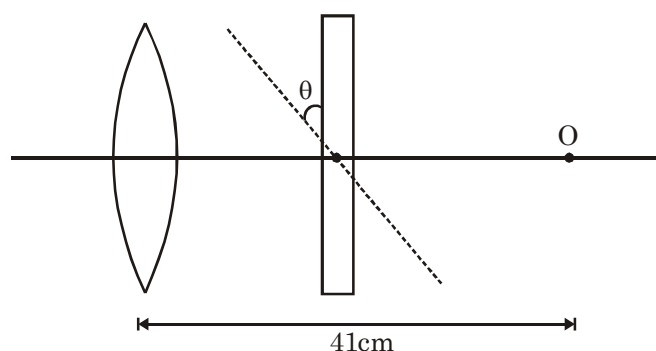
$$y_2 = 15 \sin (3\pi t + 5\pi x) \text{ cm}$$

progressing along X-axis, where x is in cm and t in second. Find the speed (in cm/sec) of the particle at

$$t = 10 \frac{1}{9} \text{ s}, x = \frac{1}{3} \text{ cm}.$$

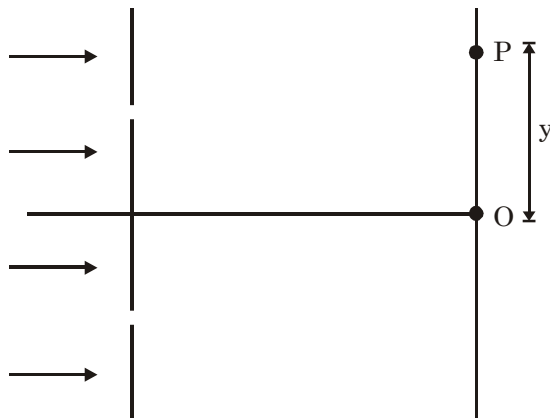
4. A point object O is placed at a distance of 41 cm from a convex lens of focal length $f = 20$ cm on its principal axis. A glass slab of thickness 3 cm and refractive index $\mu = 1.5$ is placed between the lens and the object with its faces perpendicular to the principal axis of the lens. Image of the object is formed at point I_1 . Now the glass slab is tilted by an angle of $\theta = 1^\circ$ (as shown in the figure) and the final image is formed at I_2 . Calculate the distance (in mm) between points

I_1 and I_2 . (Take $\frac{\pi}{180} = 0.17$). Consider only paraxial rays for the lens and near normal incidence for the glass slab.

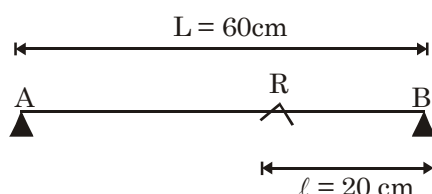


Space for Rough Work

5. In young's double slit experiment, when the slit plane is illuminated with light of wavelength λ_1 , it was observed that point P is closest point from central maximum O, where intensity was 75% the intensity at O. When the light of wavelength λ_2 is used, point P happens to be the nearest point from O where intensity is 50% of that at O. Find the ratio $\frac{\lambda_1}{\lambda_2}$.

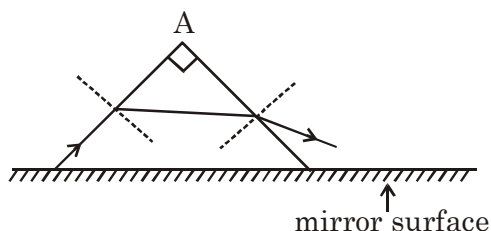


6. A string has linear mass density $\mu = 0.1 \text{ kg/m}$. A $L = 60 \text{ cm}$ segment of the string is clamped at A and B and is kept under a tension of $T = 160 \text{ N}$ [The tension providing arrangement has not been shown in the figure]. A small paper rider is placed on the string at point R such that $BR = 20 \text{ cm}$. The string is set into vibrations using a tuning fork of frequency f . Calculate all values of f below 1000 Hz for which the rider will have maximum oscillation amplitude among all points on the string.



Space for Rough Work

7. A light ray grazes the right angled isosceles prism as shown and suffers deviation of 45° . If light ray is passed in such a way that it suffers minimum deviation then what will be its total deviation in degree after refraction and reflection from horizontal mirrored surface.



8. A fish rising vertically to the surface of water in a lake uniformly at the rate of 3 m/s observes a kingfisher (bird) diving vertically towards the water at a rate of 9 m/s vertically above it. If the refractive index of water is $(4/3)$, find the actual velocity (in m/s) of the dive of the bird.

Space for Rough Work

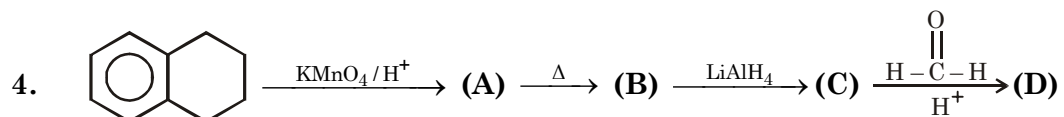
PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 40)

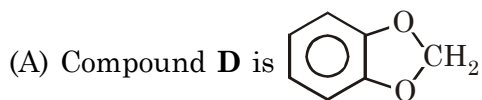
- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following pair both complex ion is low spin as well as diamagnetic.
 (A) $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{4-}$ (B) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 (C) $[\text{Cr}(\text{NH}_3)_6]^{3+}$, $[\text{Ni}(\text{en})_3]^{+2}$ (D) $[\text{Ni}(\text{en})_3]^{2+}$, $[\text{Co}(\text{H}_2\text{O})_3\text{F}_3]$
2. Which reaction(s) produce neutral oxide (nitrogen containing)
 (A) Heating of NH_4NO_2 (B) $\text{Au} + \text{dil. HNO}_3$
 (C) $\text{Cu} + \text{dil. HNO}_3$ (D) $\text{Zn} + \text{dil. HNO}_3$
3. In which of the following sp^3 hybridized boron atom is present?
 (A) $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$ (B) $\text{Na}_3[\text{B}_3\text{O}_6]$
 (C) $\text{H}_6\text{B}_2\text{O}_7^{2-}$ (D) $\text{Na}_2[\text{B}_2(\text{O}_2)_2(\text{OH})_4]$

Space for Rough Work



For the above reaction sequence correct statement(s) is/are :

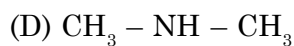
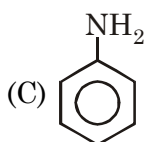
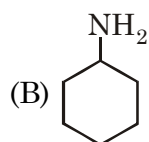
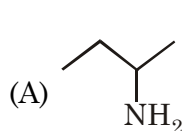


(B) Compound **C** gives red colour victor meyer test

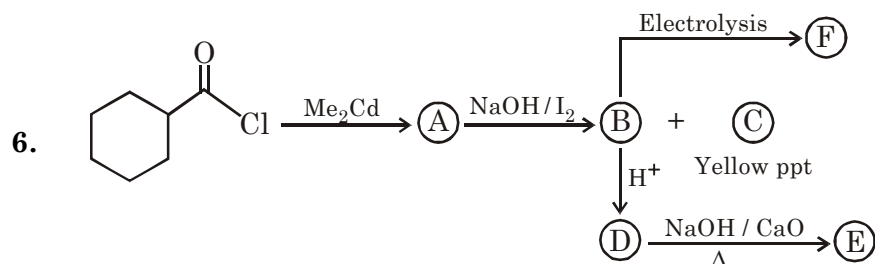
(C) Compound **A** gives brisk effervescence on treatment with NaHCO_3

(D) Compound **D** is a cyclic acetal

5. Which of the following can show positive carbyl amine test?



Space for Rough Work



Correct statement is/are :

(A) Compound F is 

(B) Compound E is 

(C) A is acetophenone

(D) In reaction from A to B, 3 equivalent of I_2 consumed, with respect to 1 equivalent of A.

7. Which is the correct statement about face centre cubic unit cell of edge length 200 pm ?

(A) Packing fraction of unit cell is 0.74

(B) The radius of largest sphere that can be fitted into voids is 51.75 pm

(C) The distance of nearest neighbours is 141.44 pm

(D) The distance of next nearest neighbours is 200 pm

Space for Rough Work

8. Blood cells in the human body have semipermeable membrane & depending upon concentration of solution inside blood cells & outside (in the blood); " Lysis" [expansion of blood cells] & " Crenatin"[Contraction of blood cells] may occur. Kidneys are responsible for keeping solution inside blood cell & blood at same concentration. Identify the options which is /are correct?
- (A) Lysis will occur when blood cells are kept in a solution which is isotonic with blood
- (B) Crenatin will occur when blood cells are kept in a solution which is hypertonic with blood.
- (C) Blood cells will have normal shape when placed in a solution isotonic with blood.
- (D) Lysis occurs when blood cells are kept in hypotonic solution as compared to blood.
9. Select the correct statement(s)-
- (A) For a spontaneous process, gibb's free energy must decrease.
- (B) Temperature is the measure of average chaotic motion of particles in the system.
- (C) On mixing two ideal gases at constant T & P, entropy increases.
- (D) On mixing two ideal gases at constant T & P, gibbs free energy decreases.
10. Select the correct match :
- | | |
|---------------------------------------|-------------------------------------|
| (A) AlN $\rightarrow$ Ionic solid | (B) HCl $\rightarrow$ Ionic solid |
| (C) NaCl $\rightarrow$ Covalent solid | (D) Si $\rightarrow$ Covalent solid |

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

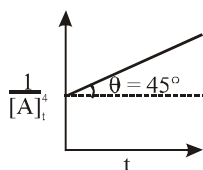
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. For the reaction : $2A \rightarrow 3B$, following graph was obtained.



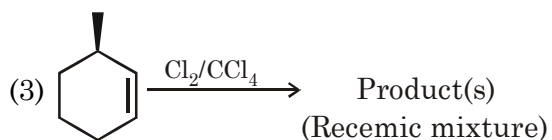
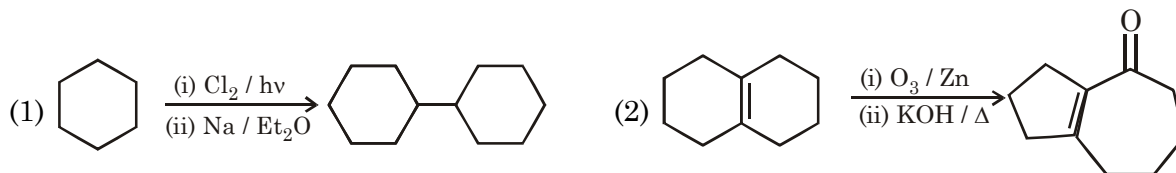
where $[A]_t$ is concentration of A at a time t [in M] & t is in sec.

From the graph, calculate rate of reaction $M \text{ sec}^{-1}$ at a concentration of $[A] = 2M$.

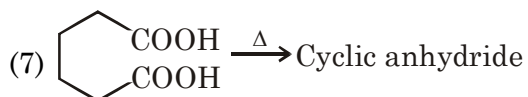
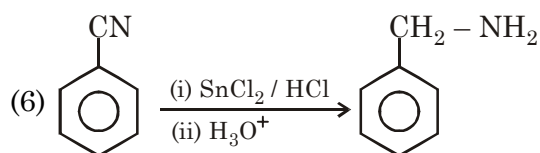
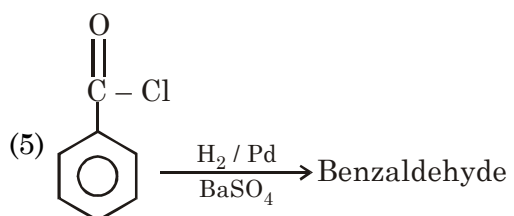
2. 0.35 molal aqueous solution of Na_xA has freezing point of $-3.255^\circ C$. If K_f of water is $1.86 K \text{ kg mol}^{-1}$, the value of x is : (Assume complete ionisation of salt Na_xA)

Space for Rough Work

3. Find number of reaction correctly matched, with their major product (or statement written with them) :

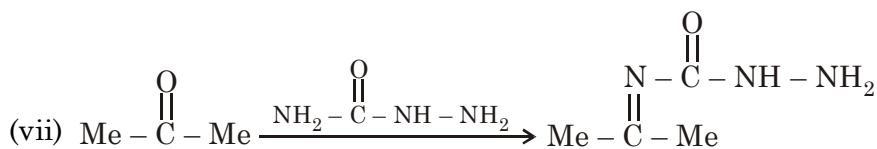
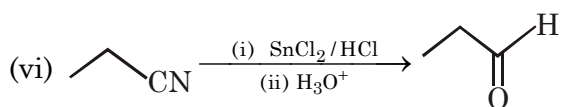
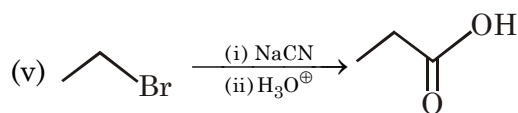
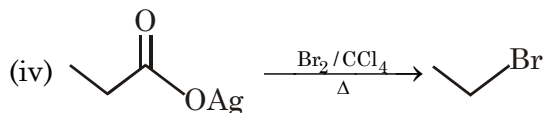
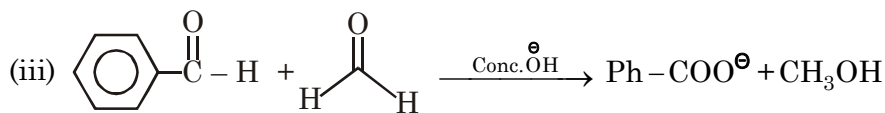
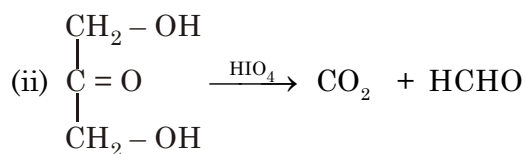
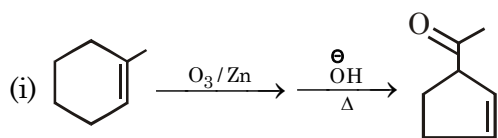


- (4) CH_3CHO gives positive Tollen's test as well as Lucas test



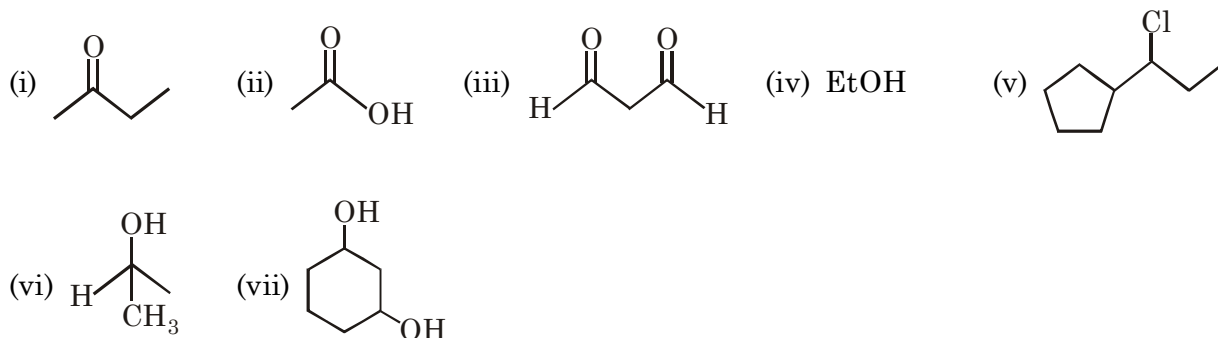
Space for Rough Work

4. Find number of reaction which are correctly matched with their major product?



Space for Rough Work

5. Find number of compounds which give yellow ppt with NaOH / I₂ :



6. Find the number of elements which are inert towards reaction with H₂O at room temperature. Be, Na, Ca, K, Ag, Li, Ba, Rb, Cs, Sr

7. Find sum of the number of atoms in the specified products of each molecule in given reactions.

(a) $K_{(s)} + O_{2(Excess)} \longrightarrow$ Principal combustion product

(b) $KCl_{(Aq.)} + \text{hexachloridoplatinic(IV) acid} \longrightarrow$ Precipitate formed in the reaction

(c) $MnSO_{4(aq.)} + \text{sodium peroxydisulphate} \xrightarrow{\text{acid}}$ "Mn" containing ion

(d) $Na_2O_{2(s)} + CO_{2(g)} \longrightarrow$ gaseous product of reaction

8. Number of reagents which can distinguish $Ag_{aq.}^{+}$ & $Pb_{aq.}^{+2}$:-

NaOH _(aq.) ,	Excess NH ₄ OH,	dil. HCl,
K ₂ CrO _{4(aq.)} ,	Excess KCN,	H ₂ S,
Aqueous KI followed by heating,	Na ₂ HPO _{4(aq.)}	

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I : (Maximum Marks: 40)**

- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. OABC is a tetrahedron of volume $\frac{2}{3}$ cubic unit. Also $BC = \sqrt{2}$, $\angle AOB = \frac{\pi}{4}$ and $OA + OB = 4$, then
 (A) area of triangle AOB cannot exceed $\sqrt{2}$ sq. unit
 (B) the length of altitude drawn from vertex C to the opposite base is $\sqrt{2}$ units.
 (C) $OC = 2$ unit
 (D) $AC > (2\sqrt{2} - 1)$
2. A and B are non singular square matrices of same order with real entries such that $\text{tr}(AA^T - AB^T A^T) = \text{tr}(ABA^T - ABB^T A^T)$, then which of the following is / are true :
 (A) $\text{tr}(A - AB) \neq 0$
 (B) $A^T = B^T A^T$
 (C) Sum of all element of matrix $A - AB$ is an odd number
 (D) B must be an identity matrix if A is invertible matrix

Space for Rough Work

3. The equation of plane passes through origin and which is equally inclined to the lines $\frac{x-1}{2} = \frac{y+1}{-2} = \frac{z+2}{-1}$ and $\frac{x-3}{8} = \frac{y-2}{1} = \frac{z-1}{-4}$ is/are -
- (A) $x + 5y - 2z = 0$ (B) $14x - 5y - 7z = 0$
 (C) $3x + 4y + 8z = 0$ (D) $2x + 7y - z = 0$
4. A parabola touches x-axis and the line $y = x$ respectively at points (1, 0) and (1, 1), then
- (A) slope of directrix of the parabola is -2
 (B) slope of axis of the parabola is $\frac{1}{3}$
 (C) focus of the parabola is at the point $\left(\frac{2}{5}, \frac{4}{5}\right)$
 (D) focus of the parabola is at the point $\left(\frac{3}{5}, \frac{1}{5}\right)$
5. Consider parabola $y^2 = 4x$. Then locus of centroid of a triangle formed by vertex of parabola and a variable focal chord is a conic
- (A) whose eccentricity is $\sqrt{2}$ (B) whose eccentricity is 1
 (C) whose length of latus rectum is $\frac{3}{4}$ (D) whose length of latus rectum is $\frac{4}{3}$
6. Let 'P' be a point on $y^2 = 4x$ which is at shortest distance from centre of circle $x^2 + y^2 - 4x - 16y + 64 = 0$ and Q is point of intersection of circle and line segment joining point P and centre of circle then
- (A) slope of tangent of circle at point Q is equal to $\left(\frac{1}{2}\right)$.
 (B) distance between centre of circle and point P is $2\sqrt{5}$.
 (C) co-ordinates of point P is (4, 4).
 (D) x-intercept made by normal at point P is 4.

Space for Rough Work

7. Let $f(x) = |x-1| ([x] - \{-x\}) \forall x \in \mathbb{R}$ where $[.]$ denotes greatest integer function and $\{.\}$ denotes fractional part function.
 (A) $f(x)$ is discontinuous function at every integer
 (B) $f(x) = -1$ can not have any solution
 (C) Range of $f(x)$ for $x \in [-1, 2]$ is $(-4, 1) \cup \{-2, 2\}$
 (D) $f'(1^+) = f'(1^-) = 0$
8. For the function $f(x) = e^x - \tan^{-1} x - 3$ which of the following holds good ?
 (A) $f(x) = 0$ has exactly one positive real root.
 (B) $y = f(x)$ has exactly one minima at negative x
 (C) $y = f(x)$ has exactly one point of minima at positive x
 (D) $y = f(x)$ is strictly increasing function.
9. If a curve passing through $P(4, 2)$ and satisfies the differential equation $\frac{dy}{dx} = \frac{y}{x - y^2}$ is a conic then which of the following is correct ?
 (A) focus of conic lies on line $y = 2$ (B) equation of directrix of conic is $4x - 17 = 0$
 (C) eccentricity of conic is 1 (D) eccentricity of conic is 3
10. Let $f : \mathbb{R} \rightarrow (0, \infty)$ in a continuous function such that $f(x) + \frac{1}{f(x)} = e^x + e^{-x} \forall x \in \mathbb{R}$, then
 (A) number of such functions $f(x)$ is 1.
 (B) number of such functions $f(x)$ is 2.
 (C) area bounded by $y = f(x)$, x -axis, ordinate $x = -1$ and ordinate $x = 1$ can be $e - \frac{1}{e}$
 (D) area bounded by $y = f(x)$, x -axis, ordinate $x = -1$ and ordinate $x = 1$ can be $2e - 2$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

										+	
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

										-	
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Let P be an interior point of $\triangle ABC$ such that $\overrightarrow{PA} + 2\overrightarrow{PB} + 3\overrightarrow{PC} = \vec{0}$, then ratio of area of $\triangle ABC$ to $\triangle APC$ is

- If $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{bmatrix}$, then sum of all possible distinct values of λ for which $AX = \lambda X$, where X is a non-zero column matrix, is

- Let L is distance between two parallel normals of $\frac{x^2}{25} + \frac{y^2}{16} = 1$, then maximum value of L is -

Space for Rough Work

4. For all admissible values of the parameter 'a' the straight line $2ax + y\sqrt{1-a^2} = 1$ will touch an ellipse whose eccentricity is equal to-
5. Slope of normal to the ellipse at a point P is $\frac{3}{4}$ and eccentricity of ellipse is $\frac{1}{3}$. If this normal makes acute angle ' β ' with its focal chord through P, then $\sin\beta$ is
6. The value of $\lim_{\theta \rightarrow 0^+} (\sin\theta)^{(\sin\theta - \sin^2\theta)}$ is equal to
7. If $\frac{d^3y}{dx^3} = -\left(\frac{dy}{dx}\right)^2 \left[k \left(\frac{d^2x}{dy^2} \right) \cdot \left(\frac{d^2y}{dx^2} \right) + \left(\frac{dy}{dx} \right)^2 \left(\frac{d^3x}{dy^3} \right) \right]$, then k is equal to
8. Let 'f' be continuous periodic and bounded function with period 3 such that $\int_0^3 f(t) dt = 6$.

Also, $F(x) = \int_0^x f(t) dt$. The value of $\lim_{x \rightarrow 0} xF\left(\frac{1}{x}\right)$ is equal to

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I	One or more correct option(s)	10	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	40
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator